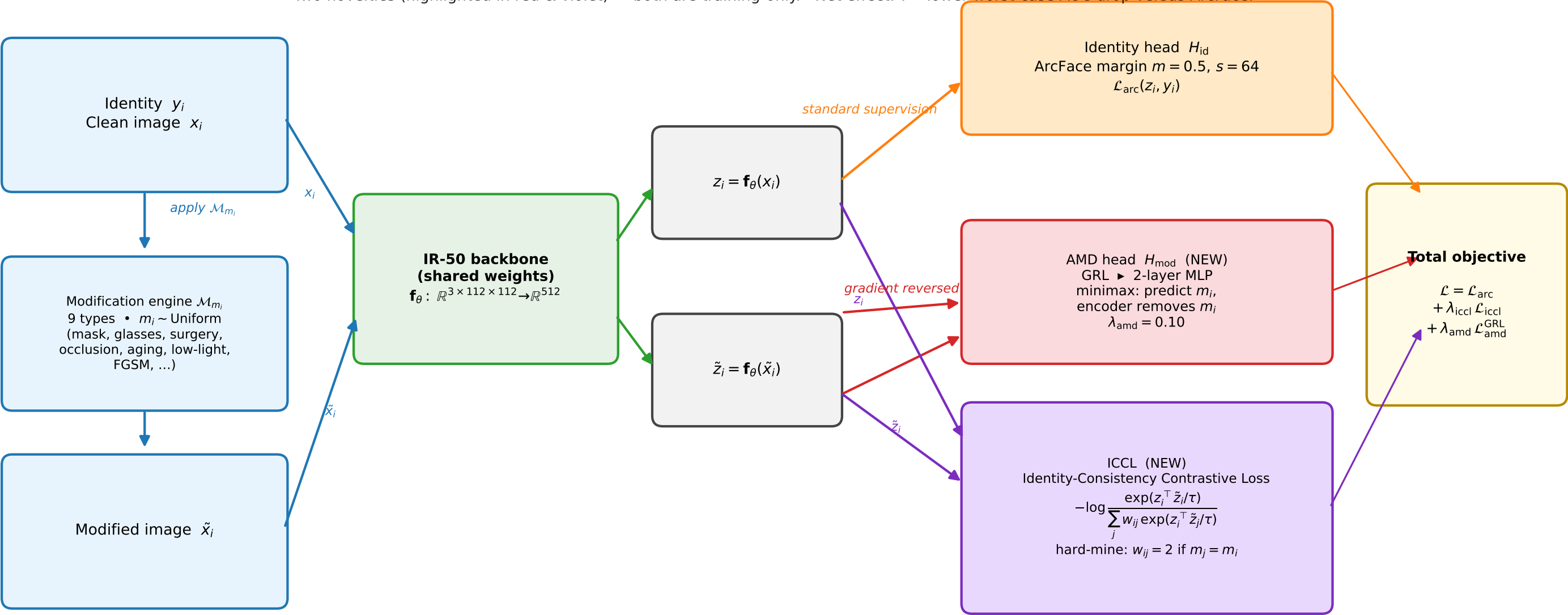


MDIE — Modification-Disentangled Identity Encoder

Training-time-only adversarial disentanglement + identity-consistency contrastive learning. Inference path is identical to ArcFace.

Two novelties (highlighted in red & violet) — both are training-only. Net effect: 7× lower worst-case AUC drop versus ArcFace.



Inference (test time): $z = \mathbf{f}_\theta(\text{image}) \rightarrow$ cosine similarity. Identical to ArcFace. No modification label is ever required.